Who Proposes, Decides: Region Proposal Governs Reference-Based Anomaly Detection with Vision–Language Judges in Tram CCTV

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Abstract

Detecting persistent appearance change inside public-transport vehicles — forgotten objects, litter, graffiti, damage — is a task that falls between industrial anomaly detection, clip-level video anomaly detection, and classical abandoned-object detection, and no public benchmark combines its three defining conditions: a fixed interior viewpoint, a per-camera clean reference, and an open set of anomaly classes. We present a fully local pipeline that proposes candidate regions against a per-camera clean reference and asks a 9B-parameter vision–language model a single yes/no question about each proposed region, and, because the two stages are independently selectable, we sweep them one at a time. Under a frozen judge, replacing a photometric proposal stage with a DINOv2 feature-based one raises frame-level F_1 from 0.931 to 0.984 and region precision from 0.694 to 0.896 while object recall does not move at all (55/73 in every arm), whereas twelve model $\times$ prompt arms fail to improve on the shipped judge at usable specificity — the spatial proposal stage, not the language model, sets the accuracy ceiling. On CDnet 2014 the same proposal stage recovers 77–100 % of annotated foreground across seven perturbation categories, and we report the two categories where a static reference fails, together with the deployment cost of the per-camera models the strongest variant requires.

Keywords: anomaly detection; vision–language models; region proposal; public transport surveillance; abandoned object detection; DINOv2; change detection

1 Introduction

Public-transport operators inspect vehicle interiors for objects left behind, litter, graffiti and damage. On a tram, the inspection happens at the end of service, by eye, and it scales with the number of vehicles rather than with the number of incidents. Interior CCTV is already installed for other reasons, so the appealing idea is to have it flag the carriages that are worth walking through. What makes the automated version of this task specific is not the anomaly classes but the observation geometry: the camera never moves, the scene is rigid, and a clean state of that exact viewpoint is available. A detector therefore never has to learn what a tram looks like in general — it only has to notice that this camera's scene differs from this camera's clean reference.

That framing puts the task between three literatures without belonging to any of them. Industrial anomaly detection [1, 2] assumes object-centric images of a single category and reports image-level scores; video anomaly detection [3, 4] assumes a moving or uncontrolled camera and decides which *clip* is anomalous; classical abandoned-object detection [5, 6] matches our observation geometry closely but was built around background subtraction and a closed-vocabulary classifier, with illumination change as its acknowledged failure mode [7, 8]. Vision–language models change what the last stage of such a system can be: instead of classifying a region into a fixed label set, one can *ask* about it. Our system does exactly that, locally, on a single workstation.

The contribution of this paper is not the system but a measurement made possible by how it is built. Because the region-proposal stage and the vision–language judge are independently selectable components sharing one box contract and one set of post-filters, each can be swept while the other is frozen. Doing so gives a result we did not expect: with the judge held fixed, replacing a photometric proposal stage by a DINOv2 feature-based one raises frame-level F_1 from 0.931 to 0.984 and region precision from 0.694 to 0.896, while object recall does not move by a single instance (55/73 in all four arms). With the boxes held fixed instead, twelve model $\times$ prompt arms fail to improve on the shipped judge at usable specificity. In this pipeline the binding constraint is spatial, not linguistic: *the proposal stage sets the accuracy ceiling, and the judge decides where under that ceiling the system sits.* To our knowledge this statement cannot be made — or even posed — in the closest training-free vision–language anomaly pipelines: LAVAD [9], VERA [10] and AnomalyRuler [11] all caption or score whole frames and segments and decide *temporally*, so none of them contains a spatial proposal stage to vary.

A companion result concerns prompting. Across our sweep, a prompt is a real lever exactly where a model is miscalibrated — on an under-firing model it moves instance recall from 39/73 to 48/73 at unchanged specificity — and no lever at all on a model that is already calibrated. Read together with the finding of Borodin et al. [12] that prompt sensitivity drops sharply *after* parameter-efficient fine-tuning, the honest joint statement is that prompting is calibration rather than capability, and that the next lever for this class of system is adaptation, not further prompt engineering. We state this as a limit of our setting, not as a general law: VERA [10] treats prompts as learnable parameters and gains from doing so.

Contributions.

1. A fully local, reference-based anomaly-detection pipeline for in-vehicle CCTV in which the region-proposal stage and the vision–language judge are independently selectable components over a shared box contract (Section 3).

2. A controlled two-stage sweep — four proposal stages under a frozen judge, twelve judges over a frozen set of boxes — showing that the proposal stage sets the accuracy ceiling while object recall stays invariant (Section 5).

3. Evidence that prompt and model selection behave as calibration and are exhausted at this scale without adaptation, stated jointly with the post-fine-tuning prompt insensitivity reported in [12].

4. An evaluation of the proposal stage on public data — CDnet 2014 [13], seven perturbation categories — so that the component carrying our main claim is verifiable outside the private dataset, including the two categories in which it fails.

2 Related Work

2.1 Anomaly detection without language

The industrial line of work defines the modern setting: train on nominal images only, score anomalies at image and pixel level, and evaluate on MVTec AD [1]. Its strongest representatives are memory-based — PatchCore [2] stores a coreset of nominal patch features and scores by nearest-neighbour distance — distribution-based — PaDiM [14] fits a per-position Gaussian over pretrained features — or distillation-based, where EfficientAD [15] pushes latency to the millisecond range. The anomalib library [16] makes these directly comparable, and we use its reference implementations as baselines (Section 5.3). Two properties of this family matter here. First, it is object-centric: images are crops of one category, not scenes. Second, it produces a score map, not a region proposal with a semantic verdict, so it answers "how anomalous" rather than "what is this".

Multi-class unified models remove the one-model-per-category assumption. Dinomaly [17] reconstructs frozen DINOv2 [18] features through a deliberately weakened decoder — a noisy bottleneck, linear attention, and a loose reconstruction constraint — so that the model cannot learn to reproduce anomalies it has never seen, and reports that a single model matches per-category specialists. Its recent successor [19] adds context-aware recentering of the reconstruction targets among other changes. We re-implemented both and report what they do on our cameras in Section 5.5, including a transfer result that this literature does not measure, because its benchmarks have no notion of a *viewpoint* that a model could fail to transfer across.

Closest to our proposal stage is AnomalyDINO [20], which shows that frozen DINOv2 patch features with a nearest-neighbour memory bank are a strong training-free anomaly detector and, notably, that this vision-only approach outperforms vision–language alternatives in the few-shot regime. Our proposal stage adopts its scoring idea and specialises it to a fixed camera (Section 3.4).

2.2 Change detection and abandoned objects

Comparing a frame against a nominal reference is change detection, and CDnet 2014 [13] is its reference benchmark: eleven categories of perturbation — shadows, camera jitter, night, dynamic background, intermittent object motion — with per-pixel ground truth and a fixed metric suite. Its `intermittentObjectMotion` category is our task almost word for word, and we use CDnet to evaluate the proposal stage in isolation (Section 5.4).

Abandoned-object detection (AOD) is the task whose *semantics* match ours. Luna et al. [5] survey the field and propose a unified evaluation protocol; the representative modern pipeline still combines dual background models with a CNN classifier [7, 8], and recent benchmarks [21] stay outdoors and road-side. Two things stand out for our purposes. The architecture is unchanged since the AVSS era [6] — background subtraction proposes, a closed-vocabulary classifier decides — and illumination change remains the openly stated failure mode. Our pipeline replaces the classifier with an open-vocabulary judge and the illumination problem with an explicit occlusion mask plus a feature-space comparison that is largely exposure-invariant.

2.3 Language-guided anomaly detection

One family puts language inside the scoring function. WinCLIP [22] scores windows against handcrafted normal/anomalous prompts; AnomalyCLIP [23] learns object-agnostic prompt vectors so that one prompt set transfers across datasets. A second family uses a large vision–language model as the detector itself: AnomalyGPT [24] produces threshold-free textual verdicts and reports that such models are weak on *localized* detail — which is precisely why our judge is never asked to localize. The survey of Ren et al. [25] organises the field into foundation models used as encoder, as detector, or as interpreter, and names prompt sensitivity as a recurring obstacle. Our judge is an *interpreter* in that taxonomy: it never produces the box.

2.4 Video anomaly detection and training-free VLM pipelines

The weakly supervised mainstream, anchored by UCF-Crime [3] and surveyed in [4, 26], learns from video-level labels and outputs a per-frame anomaly score. It is not our benchmark: its anomalies are events with temporal extent, ours are static changes that persist indefinitely, and its cameras are uncontrolled.

Three recent systems are our closest competitors because they, like us, use a vision–language model without task-specific training. LAVAD [9] captions frames with an off-the-shelf captioner, cleans the captions with an LLM, and scores temporally. VERA [10] learns *verbalized* guiding questions as parameters and aggregates segment scores into frame scores. AnomalyRuler [11] induces normality rules from a few normal frames and applies them as language. All three decide *when* something is anomalous, and all three consume *whole frames*. AnomalyRuler is explicit about it: its visual perception module "utilizes a VLM to convert video frames into text descriptions", prompted with *"What are people doing? What are in the images other than people?"* — a question about the frame, not about a region of it. None of the three contains a spatial region-proposal stage that could be varied while the language model is held fixed, which is why the experiment of Section 5.1 has no counterpart in this line of work.

2.5 Can the judge be trusted?

Our judge answers a binary question about a crop, which is exactly the probing format of POPE [27]: POPE shows that state-of-the-art vision–language models hallucinate objects at high rates under yes/no probing, and that the measured rate depends strongly on how the negative examples are sampled. That is the reason we report every recall next to the judge's YES rate (Section 4.2): a model that answers YES to most crops recovers most instances mechanically, and calling that detection would be an artefact of the sampling. The broader concern of using a model as an evaluator is by now a named methodological problem [28].

2.6 Region proposal and in-vehicle CCTV

Open-vocabulary detectors — Grounding DINO [29], YOLO-World [30] — propose boxes from text and are an obvious alternative proposal stage; SAHI [31] shows that tiled inference substantially improves small-object detection, which is the regime a forgotten phone falls into at CCTV resolution.

Closest to our optional veto stage is BEM [32], a training-free, weight-frozen module attached to a pretrained detector at inference: it estimates a background embedding from recent unlabelled frames of a fixed camera, keeps a prototype memory, and re-scores the detector's logits with an inverse-similarity penalty, reducing false positives while preserving recall. Its premise is ours — on a fixed camera the quasi-static background is a stable, label-free prior — and its position in a pipeline is ours too: it acts *after* a proposal stage and can only suppress. It is therefore independent corroboration of one half of our finding (Section 5.1), from a different task and a different detector family. Two things separate it from our work: its prior is a single global frame embedding rather than a per-position reference, and it re-scores the outputs of a closed-vocabulary detector, so what it can never do is put a box around something the detector did not already propose.

The in-vehicle deployment domain itself is thinly covered, and the review by Caetano et al. [33] says so in terms that match our setting closely. It identifies the extension of outdoor abandoned-object and abnormal-behaviour work to in-vehicle monitoring as "a relevant research opportunity that has been overlooked in the accessible literature", names moving backgrounds and frequent illumination changes as the specific issues previous surveys ignore, and reports that work in this domain remains "fully dependent on the availability of private datasets". It also states the operational priority — the identification of unattended objects as the main safety challenge in public transport — and the very heuristic our person veto implements: an object counts as abandoned when its presence is not expected *without a person present in the frame*. Its proposed remedy for the data scarcity is synthetic augmentation, which is the family our inpainted cases belong to (Section 4.1). On the benchmark side, Bus Violence [34] is, to our knowledge, the only public in-vehicle CCTV anomaly benchmark, and it targets violent *behaviour* — an event, with people as the subject — rather than persistent change with people as the principal nuisance.

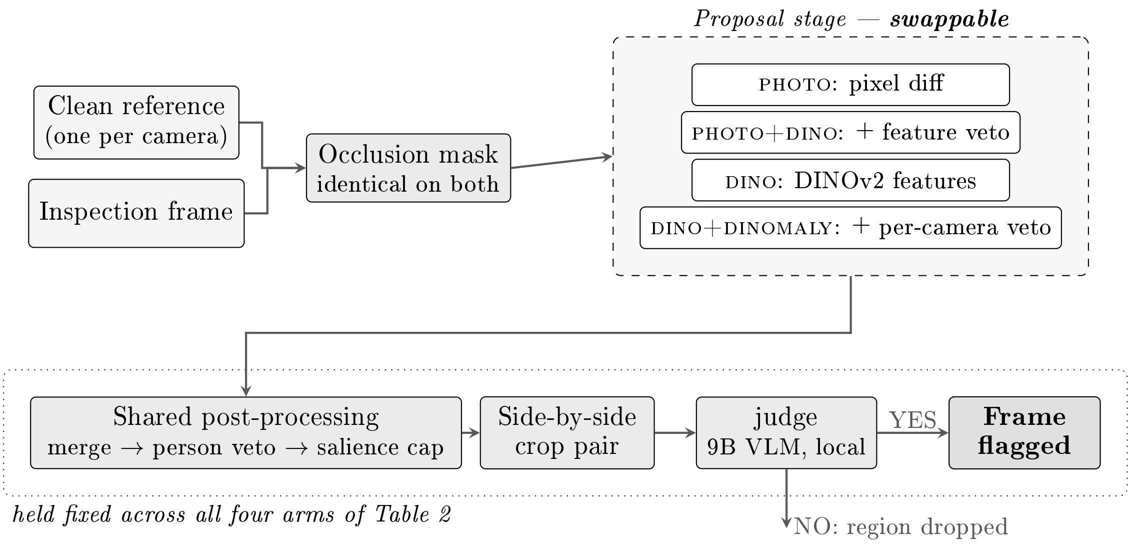

Figure 1: Overview of the two-stage pipeline. The occlusion mask is applied identically to the reference and to the inspection frame; the proposal stage emits boxes in reference pixel space; the shared post-processing (merge, person veto, salience cap) is identical for every proposal stage; the judge sees one side-by-side crop pair at a time and never the whole frame.

The gap our data occupies is therefore documented, not assumed, and it is also the reason we cannot fill it publicly (Section 6).

3 Methodology

3.1 Task and design principle

The system observes the fixed interior cameras of a tram at 1280×720 px. The scene is rigid, the viewpoint never changes, and the vehicle is emptied and inspected at the end of each service, so a clean state of each view is available. The task is to flag *persistent appearance change*: a forgotten object, litter, a graffiti tag, or damage to a fitting. People are explicitly not anomalies, and neither is anything a person is wearing, holding or sitting on — a distinction that matters more than it sounds, since a jacket on a seat is an anomaly and the same jacket on a passenger is not.

The pipeline has two stages, illustrated in Figure 1: a *proposal stage* that proposes candidate rectangles by comparing the inspection frame to a nominal reference of the same view, and a *judge* that answers a yes/no question about each proposed region. The design principle of this paper is that the two stages fail for different reasons and are improved by different work, so they are implemented as independently selectable components sharing a single contract — a list of boxes in reference pixel space — and a single set of post-filters. Swapping the proposal stage therefore changes which regions reach the judge, and nothing else, which is what makes the controlled comparison of Section 5.1 possible rather than a comparison of two different pipelines.

Everything runs locally on one workstation (NVIDIA RTX 3080 Ti, 12 GB); no frame leaves the machine. This rules out hosted vision–language APIs and caps the judge at roughly nine billion parameters in 4-bit quantisation, which is why model selection is reported as a measurement (Section 5.2) rather than assumed.

3.2 Reference and occlusion mask

A *reference* is one manually selected frame of a camera in its clean state. Windows are the dominant nuisance in a tram: exterior motion and daylight change produce large, fast and entirely uninteresting differences. They are removed by a camera-wide occlusion mask — polygons drawn once per view and filled with black — applied identically to the reference and to every inspection frame at pipeline input. Masking only one side would make the mask itself the largest detected change. Masked pixels are excluded from every downstream statistic: the photometric channels ignore them, and the feature-based stages drop the patches falling inside them before computing any scale estimate.

3.3 Proposal stage I: photometric difference (PHOTO)

Let R and I be the masked reference and inspection frames in grayscale, and G_σ a Gaussian blur of radius $\sigma = 3$ px. The base change map is

$$D(x) = \big|G_\sigma(I)(x) - G_\sigma(R)(x)\big|, \qquad M_{\text{base}}(x) = \mathbf{1}[\,D(x) > \tau_1\,], \tag{1}$$

with $\tau_1 = 40$. M_{base} is downscaled by 4, dilated by 2, and its connected components become boxes retained when their area lies in $[5 \times 10^2,\, 4 \times 10^5]$ px.

A single global threshold cannot both catch faint anomalies and keep the region count bounded: lowering τ_1 to 25 merges busy frames into single large blobs that the area gate then deletes. The base detector is therefore left untouched and two channels only *add* candidates, as a union of region lists rather than of masks — a mask union would merge the boxes back together:

- a second photometric pass at $\tau_2 = 30$, contributing at most eight boxes that overlap no base region, ranked by salience. This recovers low-contrast solid objects, such as a dark bottle on a dark floor, which appear at $\tau \in [30, 35]$ but not at τ_1;
- an *added-edge-energy* channel

$$E(x) = \operatorname{relu}\big(\|\nabla I(x)\| - \|\nabla R(x)\|\big) \cdot \mathbf{1}[\,\|\nabla R(x)\| < \tau_{\text{flat}}\,], \tag{2}$$

  blurred and thresholded at $\tau_E = 1.5$ with $\tau_{\text{flat}} = 6.0$, contributing at most four boxes. Sensor and JPEG noise is symmetric between the two frames and cancels in this one-sided difference, whereas a faint tag adds stroke edges on a flat panel. This is what reaches faint graffiti that no global photometric threshold can reach without flooding: on our single-camera benchmark the two channels lift localization recall from 41/45 to 45/45 instances, at a 52 % increase in candidate regions on anomalous frames.

3.4 Proposal stage II: DINOv2 patch features (DINO)

Both frames are resized to 1120 px wide and encoded by a frozen `dinov2_vits14_reg` backbone [18] (22 M parameters), giving an 80×45 patch grid at 24 px per patch. Let $f_I(p)$ and $f_R(q)$ denote patch embeddings and $\mathcal{N}(p)$ the ± 1-patch neighbourhood of p in the reference grid. Each patch is scored by

$$s(p) = \min_{q \in \mathcal{N}(p)} \left(1 - \frac{\langle f_I(p),\, f_R(q)\rangle}{\|f_I(p)\|\,\|f_R(q)\|}\right). \tag{3}$$

Where AnomalyDINO [20] builds one memory bank over all nominal patches — appropriate for object images that are centred but not aligned — our camera is fixed, so a patch is compared only to nominal patches at the same grid position, up to ± 1 patch of slack absorbing sub-patch jitter. That is a far tighter null hypothesis: a seat cushion is compared to *that* seat, not to any seat in the image.

A patch is flagged when it exceeds both a robust z-score and an absolute floor,

$$z(p) = \frac{s(p) - \operatorname{med}(s)}{1.4826 \cdot \operatorname{MAD}(s)}, \qquad \operatorname{flag}(p) = \mathbf{1}[\,z(p) \geq 4.0 \;\wedge\; s(p) \geq 0.08\,], \tag{4}$$

with median and MAD taken over valid (unmasked) patches only. Both conditions are necessary, and this was measured rather than assumed: the per-frame MAD varies fivefold across our cases (0.006 on a same-session pair, 0.030 on a frame containing a person), so a single z threshold corresponds to an absolute cut of 0.043 on one frame and 0.225 on another. On a genuinely near-identical pair the MAD collapses and the z amplifies feature jitter into more than eighty spurious regions; the floor removes those, since such frames sit at $p_{99} \approx 0.07$, while the z continues to handle cross-session pairs, where the whole distance map shifts upward. For scale, truly anomalous patches reach $s \in [0.37, 0.69]$ and the faintest true instance we measured peaks at $s = 0.126$. The resulting binary map is passed through the same connected-component routine as the photometric channels, so box geometry is produced identically in both families.

3.5 Proposal stage III: reconstruction veto (DINO+DINOMALY)

An optional third arm keeps the feature proposal stage and adds a per-camera model that deletes the boxes it can reconstruct. We re-implemented Dinomaly [17] — a reconstruction model over frozen DINOv2 features with a noisy bottleneck, linear attention and a loose reconstruction constraint — trained on nominal footage of that camera, and apply it as a veto at a reconstruction-error threshold of 0.05 inside each proposed box. A veto can only delete boxes, so this arm's accuracy ceiling is exactly that of the proposal stage it filters; it is included because reaching that ceiling at half the cost is itself informative, and because it requires one checkpoint per camera, a deployment cost quantified in Section 5.5. For completeness the sweep also includes PHOTO+DINO, in which the photometric diff proposes and DINOv2 features veto — an arm that shares the photometric proposal stage's boxes exactly and therefore isolates the proposer/veto distinction.

3.6 Shared post-processing

Every proposal stage feeds the same four steps. Holding them fixed is what makes Section 5.1 a controlled comparison.

1. **Merge.** One real object rarely produces one connected component: a backpack and its strap, or a phone and the sliver of seat edge beside it, land separately, and the extra channels add their own boxes on top. Boxes whose gap is below 24 px are merged when the union box is at least 50 % filled by its parts; a union emptier than that is treated as two objects and left apart. Without this step the same object reached the judge as two to five independent fragments.
2. **Person veto.** YOLOv8-nano [35] runs once per inspection frame (≈20 ms on GPU), and candidate regions whose intersection over their own area with a person box reaches 0.6 are dropped before the judge. This is what separates a worn jacket from a forgotten one, which no label blacklist can do. It was verified to lose zero ground-truth instances, and degrades to a warning if the detector is unavailable.
3. **Salience cap.** At most 25 regions per frame, ranked by salience, bounding worst-case judge cost.
4. **Crop rendering.** Each surviving box is rendered as a side-by-side pair — reference on the left, inspection on the right — with 40 px plus 75 % of the region size as context, each half upscaled to at least 320 px on its shorter side. The upscale is not cosmetic: a phone on a seat occupies about 50 px, and without it the judge dismisses small dark objects as reflections.

Regions dropped at steps 2 and 3 never reach the judge and are recorded as counts only; every region that does reach it is recorded with its verdict and its outcome, which is what allows the localization-only and end-to-end views of one run to be reconciled.

Table 1: Composition of the private evaluation set: 68 cases, 73 instances, 3 trams, 9 camera views.

Instances by type		Cases by provenance	
forgotten object	53	real camera frame	54
graffiti	7	inpainted anomaly	11
litter	7	self-staged	2
damage	6	viewpoint variant	1
total	**73**	**total**	**68**

3.7 The judge

The judge is a local vision–language model served through Ollama and asked a yes/no question about *one crop pair at a time.* It never sees the whole frame and is never asked to localize anything: the box is already drawn. This is a deliberate restriction of the language model to the operation the hallucination literature finds it most reliable at — a binary probe [27] — and it is what makes the two stages separable in the first place.

The shipped judge is a 9B GLM-V–family model [36] at temperature 0.1, 4096 context, 512 predicted tokens. The prompt states the four anomaly classes, states three explicit NO conditions (a brightness, shadow or reflection change only; a person or anything they wear, hold or sit on; scratches and glare on metal or glass), and instructs the model to answer YES only if it can name a specific new object, marking or damage. The reply is `YES`/`NO` followed by a two- to four-word name.

The named answer is used: three post-filters override a YES when the label names a person or body part, when it reports a *disappearance* rather than an appearance, or when a small-object word is attached to a region larger than 6×10^3 px. These fire only on a YES, so a rejection is never misattributed to a filter. A frame is flagged when at least one region survives.

4 Experimental Setup

4.1 Data

The private evaluation set comprises 68 labelled cases over three trams and nine camera views: 31 anomalous and 37 clean frames carrying 73 ground-truth instances. Table 1 gives the composition. Every case names the reference it is compared against, and instance boxes are expressed in that reference's pixel space.

Eleven cases carry an anomaly composited into a real frame of the same camera by image inpainting. They are labelled as such and exist because two of the four anomaly types — damage and graffiti — are rare enough in a working tram that waiting for them was not possible within the study period. Synthetic augmentation of this kind is what Caetano et al. [33] recommend for exactly this domain, where they report the lack of public in-vehicle data as the binding constraint. People are never instances: a kept box on a person counts as a false-positive region.

We state the obvious limitation once, here, and return to it in Section 6: this is a small, single-deployment, non-public dataset. Section 5.4 is the mitigation.

4.2 Metrics

Three levels are reported, because the central claim of the paper is precisely that they move independently.

Frame level. Each case is a binary decision — a frame is flagged when at least one region survives the judge. We report F_1, recall and specificity over the 68 cases.

Object level. A ground-truth instance counts as detected if any kept region overlaps it under a lenient rule (IoU > 0.1 or centre containment). A strict count at IoU ≥ 0.3 is reported alongside, and *the gap between the two is the box-quality signal* — a proposal covering most of the frame satisfies the lenient rule trivially and is useless to a judge that only sees the crop. Kept regions matching no instance are false-positive regions, and region precision is the fraction of kept regions that match one.

Image-level AUROC. For the comparisons against industrial anomaly-detection models (Section 5.3, Section 5.5), which produce score maps rather than regions, we use the standard metric of that literature,

$$\text{score}(x) = \text{mean}\big(\text{top } 1\,\%\text{ of patch scores of } x\big), \qquad \text{I-AUROC} = \Pr\big[\,\text{score}(x^-) > \text{score}(x^+)\,\big], \tag{5}$$

with x^- anomalous, x^+ clean, and masked patches excluded from the top-1 % pool.

One reporting rule. Any recall is read next to the judge's YES rate. With 73 instances among 654 proposed regions, a calibrated judge answers YES about 11 % of the time; a judge answering YES to 88 % of crops also recovers most instances, and reporting that as detection would be an artefact of the proposal distribution rather than a property of the model. This follows the sampling concern made explicit by POPE [27].

4.3 Per-camera sample sizes

The AUROC comparisons are per camera, and their sample sizes differ by an order of magnitude, so we state them once and let them bound every claim that follows. Camera `tram_1762` contributes 16 anomalous and 11 clean frames (176 pairs); the four `3333` cameras contribute 14 anomalous and 7 clean frames in total (98 pairs), of which no individual camera exceeds 8 pairs; the three `1760` cameras have no labelled anomalies and contribute score levels rather than AUROCs. Only the `tram_1762` column and the `3333`-pooled column carry enough pairs to support an AUROC claim, and individual `3333` columns are reported for completeness only.

4.4 Public-data protocol

To evaluate the proposal stage on data we did not build, we run it on CDnet 2014 [13] across seven categories. The reference is the per-pixel median of up to 50 frames preceding each sequence's `temporalROI`, which reproduces our clean reference by construction; every k-th ROI frame is then evaluated. The official label semantics (static, hard shadow, outside ROI, unknown, motion) and the spatial ROI are respected, and the ROI is resampled with nearest-neighbour interpolation where a sequence ships an ROI of different size from its frames.

The person veto is **disabled** for these runs. On CDnet the annotated foreground *is* people, so leaving it enabled would measure the veto rather than the proposal stage. Two views of each configuration are reported: the per-pixel change mask, and the rasterised proposal boxes — the latter being what actually reaches the judge — both scored with the official CDnet metrics.

We also report a qualitative check on ABODA, the one public dataset whose anomaly class is ours. It ships eleven CCTV sequences and no annotations, so it yields figures and not numbers; the event-level protocol of Luna et al. [5] would require the temporal annotation package we were unable to obtain (Section 6).

Table 2: Four proposal stages, judge held fixed. 68 cases, 73 instances, three trams, nine views. Object recall does not move; everything else does.

Proposal stage	**Regions**	**Frame F_1**	**Frame Re**	**Frame Sp**	**Obj. recall**	**Strict**	**Reg. prec.**
PHOTO	1192	0.931	0.871	1.000	55/73	0.534	0.694
PHOTO+DINO	727	0.931	0.871	1.000	55/73	0.534	0.773
DINO	1228	**0.984**	**0.968**	1.000	55/73	**0.630**	**0.896**
DINO+DINOMALY	**654**	**0.984**	**0.968**	1.000	55/73	**0.630**	**0.896**

4.5 Implementation and reproducibility

All experiments run on a single NVIDIA RTX 3080 Ti (12 GB). Models: DINOv2 ViT-S/14 with registers (frozen) [18], YOLOv8-nano for the person veto [35], a 9B GLM-V–family judge [36] in 4-bit via Ollama, and our Dinomaly re-implementation [17]. The judge sweep additionally covers Qwen2.5-VL-7B [37], Qwen3-VL-8B and InternVL3.5-8B [38]. Baselines use the anomalib 2.6 reference implementations [16].

Per-region judge verdicts are cached on a key covering image, reference, box, model, prompt and occlusion mask, so a re-run reproduces a previous run exactly while a changed mask or prompt correctly invalidates it. The cache key deliberately excludes the proposal stage: a box is a box, so identical coordinates from two proposal stages share one verdict — which is also what makes the sweep of Section 5.1 affordable.

Code, ground truth, per-run reports and a decision log are available in the project repository. The tram footage itself cannot be released, which is the reason Section 5.4 exists.

5 Results

5.1 Four proposal stages under a frozen judge

Table 2 sweeps the proposal stage with the judge held fixed.

The object-recall column is the one to read first: it does not move. Fifty-five of seventy-three instances, in all four arms, to the instance. Everything else does move — frame recall from 0.871 to 0.968, region precision from 0.694 to 0.896, kept-but-wrong regions from 30 down to 7, and strict-IoU boxing from 0.534 to 0.630 — while frame specificity stays at 1.000 throughout, so none of this is bought with false alarms.

Three consequences follow, and the third is the paper.

First, *"better boxes find more objects" is the tempting reading of Table 2 and it is false.* The localization-only view (Table 3) shows the feature proposal stage localizing four instances the photometric one misses — and the judge rejects them anyway. The proposal stage sets the ceiling; the judge decides where under it the system sits.

Second, *a veto is not a proposal stage.* Rows 1–2 of Table 2 are identical on every quality column and differ only in cost, because a veto deletes boxes and never redraws one; the same holds for rows 3–4, where DINO+DINOMALY matches DINO to the digit for 47 % fewer judge calls. Matching the ceiling means none of the 574 deleted boxes held a true instance. So the operative question is never *which filter* but *who proposes.*

Third, *the binding constraint is spatial, not linguistic.* With the judge frozen, changing the proposal stage moves frame F_1 by 0.053 and region precision by 0.202. Section 5.2 shows what changing the judge does instead.

The mechanism behind the strict/lenient gap is blunt and measurable. On tram 3333 the photometric proposal stage scores 5/26 strict against the feature proposal stage's 19/26, and its worst box covers 911 360 px of a 1280×720 frame — 98.9 % of the image. Such a box satisfies the lenient rule trivially and is worthless to a judge that only ever sees the crop. Figure 2 shows one

Table 3: Localization only, no judge calls. The lenient rule barely separates the two families; the strict rule separates them decisively.

Proposal stage	**Instances localized**	**Strict IoU**	**Regions**
PHOTO	65/73	42/73	1192
PHOTO+DINO	65/73	42/73	727
Dinomaly alone	62/73	48/73	813
DINO	**69/73**	**58/73**	1228
DINO+DINOMALY	**69/73**	**58/73**	654

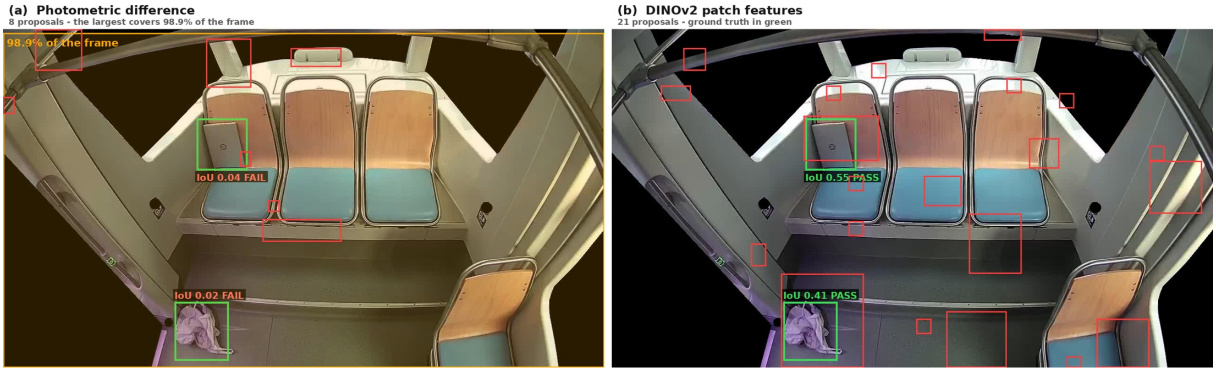

Figure 2: The same inspection frame under both proposal families. Ground truth in green, proposals in red, and the box covering 98.9 % of the frame highlighted in orange. Each ground-truth box carries the best IoU achieved and the verdict of the strict rule (IoU $\geq$ 0.3). Both instances fail under the photometric stage and pass under the feature stage.

frame where this happens, with the per-instance IoU printed on the image so the verdict can be checked without the caption: the photometric stage scores 0.042 on a laptop left on a seat and 0.016 on a bag on the floor — whose only photometric cover *is* the 98.9 % box — against 0.555 and 0.408 for the feature stage. The figure also shows what the feature proposal stage costs: 21 proposals against 8, which is why Table 2 records no saving in judge calls for DINO and why the veto arm exists at all.

5.2 Twelve judges over a frozen set of boxes

Table 4 inverts the experiment: the boxes are frozen at the 654 crops produced by DINO+DINOMALY, and four models are crossed with three prompts. The calibration anchor is that 73 instances among 654 regions imply that a calibrated judge answers YES about 11 % of the time.

No arm improves on the shipped configuration at usable specificity. That is a result and not a null result: it is the control that keeps Section 5.1 honest, since the first objection to "the proposal stage decides" is that we chose a weak judge.

Three readings follow. *The 63/73 of the Qwen arms is not detection but non-rejection*: at an 88.5 % YES rate a judge mechanically recovers almost everything handed to it, and frame specificity collapses to 0.081. This is the concrete demonstration of why every recall in this literature must be read next to the YES rate. *The prompt is a real lever only where there is slack*: on InternVL3.5, which under-fires at 6.9 %, the balanced prompt moves 39/73 to 48/73 at specificity 1.000 unchanged; on the GLM-V judge, already calibrated at 10.2 %, the same prompt gains zero instances and costs specificity. *Therefore prompting is calibration rather than capability* — at this scale, for this set of models, and without adaptation. All three qualifiers are necessary: VERA [10] treats prompts as learnable parameters and gains from it, and Borodin et

Table 4: Judge sweep on a frozen set of 654 crops: four vision–language models $\times$ three prompts, 2 h 07 of GPU. Nothing improves on the shipped configuration at usable specificity.

Model	**Prompt**	**YES %**	**Instances**	**Frame F_1**	**Frame Sp**	**Reg. prec.**
GLM-V 9B (Flash)	conservative	10.2	**55/73**	**0.984**	**1.000**	0.896
GLM-V 9B (Flash)	balanced	10.4	55/73	0.968	0.973	0.882
GLM-V 9B (Flash)	lenient	13.6	56/73	0.909	0.865	0.719
InternVL3.5-8B	balanced	8.7	48/73	0.984	1.000	0.912
InternVL3.5-8B	lenient	7.8	43/73	0.931	1.000	0.922
InternVL3.5-8B	conservative	6.9	39/73	0.893	1.000	0.933
Qwen3-VL-8B	lenient	28.9	63/73	0.690	0.297	0.376
Qwen3-VL-8B	balanced	40.5	63/73	0.659	0.189	0.268
Qwen3-VL-8B	conservative	39.6	63/73	0.645	0.135	0.274
Qwen2.5-VL-7B	balanced	54.3	64/73	0.652	0.162	0.200
Qwen2.5-VL-7B	conservative	48.3	63/73	0.645	0.135	0.222
Qwen2.5-VL-7B	lenient	88.5	63/73	0.632	0.081	0.123

Table 5: Image-level AUROC per camera, own-camera models. Only the `tram_1762` column carries enough pairs to support a claim (Section 4.3).

Method	**3333-c52**	**3333-c53**	**3333-c54**	**3333-c55**	**tram_1762**
	2 pairs	*8 pairs*	*8 pairs*	*8 pairs*	*176 pairs*
PaDiM [14]	1.000	0.875	1.000	1.000	**0.915**
PatchCore [2]	1.000	0.625	1.000	1.000	**0.841**
Dinomaly [17] *(anomalib)*	1.000	1.000	1.000	1.000	**0.886**
AnomalyDINO [20] *(anomalib)*	1.000	0.750	1.000	1.000	**0.835**
Dinomaly *(ours)*	1.000	0.750	1.000	1.000	**0.920**
Dinomaly2 *(ours)*	1.000	1.000	1.000	1.000	**0.801**

al. [12] report that prompt sensitivity drops sharply after parameter-efficient fine-tuning. Our result and theirs are two halves of one finding, and the joint statement is that the next lever for this class of system is adaptation, not further prompt engineering.

5.3 Are the learned baselines enough?

Table 5 runs anomalib 2.6 reference implementations on the same cameras, each model fitted on that camera's nominal frames and tested on that camera's cases. Scores are each method's native image score; I-AUROC is rank-based, so the comparison is valid despite differing score definitions.

Two things fall out, and the second matters more than the table. Our re-implementation is sound: it matches the anomalib reference on four of five cameras, closes the fifth with the Dinomaly2 variant, and on the only column with real statistics sits slightly above the reference (0.920 against 0.886). A re-implementation cannot be reported without such a check, and it passes.

More importantly, *frame-level anomaly detection on this data is not the hard part.* Every standard method, untuned, lands between 0.835 and 0.920 on the 176-pair column and saturates at 1.000 on three of the four remaining views. "Does this frame contain an anomaly" is close to solved by a 2021 memory-bank baseline that takes nine seconds to fit. This is why the claims

Table 6: CDnet 2014, photometric proposal stage, pooled over all evaluated frames. *Mask* is the per-pixel change mask; *boxes* is the rasterised proposals, which is what reaches the judge. Read the two families of column separately: the stage is a recall-first proposer by design.

Category	**Seq.**	**Frames**	**Mask**				**Boxes**	
			F_1	Re	Pr	FPR	Re	Pr
`badWeather`	4	899	**0.675**	0.557	0.856	0.0014	0.918	0.095
`cameraJitter`	4	159	0.566	0.647	0.504	0.0266	0.924	0.105
`shadow`	6	710	0.496	0.618	0.415	0.0463	0.942	0.178
`nightVideos`	6	619	0.434	0.466	0.406	0.0100	0.886	0.129
`intermittentObjectMotion`	6	1216	0.406	0.487	0.349	0.0350	0.812	0.139
`dynamicBackground`	6	667	0.234	0.808	0.137	0.0714	0.998	0.028
`lowFramerate`	4	264	**0.207**	0.278	0.164	0.0277	0.767	0.052

of this paper are made at region and verdict level: *which* region, and *what* it is — a language verdict a human can read, which no row of Table 5 produces — and at what operating point that holds.

5.4 The proposal stage on public data

Table 6 evaluates the photometric proposal stage on seven CDnet 2014 categories, without the judge in the loop, so that a pipeline failure can be attributed: the region was never proposed, or it was proposed and misjudged.

The category that is our task. CDnet's `intermittentObjectMotion` is defined by its authors as videos containing background objects moving away, abandoned objects, and objects stopping for a while before moving away — our task, almost word for word. Per sequence, the mask F_1 ranges from 0.978 on `streetLight` — a fixed camera, stable light, a car that stops, a score that would be competitive on the CDnet leaderboard — down to 0.143 on `winterDriveway` (snow, changing daylight) and 0.138 on `tramstop` (outdoor light drift). The stage is excellent exactly while its reference stays valid and collapses when it does not, and CDnet finds that failure mode in one afternoon.

Read recall and precision separately. Across all seven categories the boxes recall 0.767–0.998 while their precision stays between 0.028 and 0.178. This is by design, not by accident: our proposal stage is a recall-first proposer whose false positives are meant to be removed downstream by a judge that CDnet does not run. Reporting only F_1 would say the stage is weak; reporting both columns says what is true — it rarely misses, and it over-proposes on purpose. It should also be noted that the boxes row is scored against pixel-accurate ground truth by a method that emits rectangles, which structurally caps its precision.

Which of our assumptions are load-bearing. Read top to bottom, Table 6 ranks them. `badWeather` (0.675) and `cameraJitter` (0.566) are the good news and are the two categories that matter most for a tram: snow and rain against a median reference cost almost nothing (FPR 0.0014), and camera shake — a tram vibrates continuously — is survivable. `dynamicBackground` (0.234) and `lowFramerate` (0.207) are failures, and both lie outside our deployment: moving water and foliage do not exist inside a vehicle, and we control the sampling rate. `shadow` (0.496) and `nightVideos` (0.434) are the honest middle, and these two *are* ours: a tram runs at night and through moving shade. That is the number to quote when asked what the proposal stage does under illumination change, and it is mediocre.

Table 7: Cross-camera ablation. Context-aware recentering is the only element that touches the margin — and it is also what costs the home camera.

Variant	**Clean inflation** *foreign ÷ own*	**Margin collapse** *own ÷ foreign*	**Transfer** *median, pooled*	**tram_1762** *self, 176 pairs*
Dinomaly (v1)	2.13×	41×	0.367	**0.920**
Dinomaly2, CAR off	1.73×	43×	0.393	**0.903**
Dinomaly2, CAR on	**1.52**×	**10**×	0.449	**0.801**

Two claims of Section 5.1 reproduce on public data. The gated variant PHOTO+DINO scores 0.252 mean F_1 against the ungated diff's 0.254, at slightly lower recall (0.898 against 0.921) — independent confirmation, on data we did not build, that a veto is a cost lever and never a quality one. Conversely, the feature proposal stage does *not* transfer to CDnet: it reaches 0.283 mean recall and fires not at all on `tramstop`. Two explanations were tested and ruled out. Sweeping its z threshold from 4.0 to 1.5 and its floor from 0.08 to 0.02 moves recall from 0.616 to 1.000 while precision stays pinned between 0.011 and 0.022 and the false-positive rate climbs to 0.98 — there is no threshold at which it is useful; it goes from silent to covering the frame. Rebuilding the reference from a single real frame rather than a 50-frame median leaves it unchanged (F_1 0.029 against 0.028) while making the photometric stage distinctly worse, so the composite reference is not the cause. What remains is the assumption itself: the feature stage compares each patch to the same grid position of a reference from that same fixed camera, and quantises its boxes to the patch grid. That is a good trade on a tram interior, where an anomaly spans a large fraction of a patch, and a bad one on 320×240 outdoor footage scored against pixel-accurate ground truth. *Our best proposal stage is best conditionally*, and the condition is the deployment geometry.

5.5 The cost of per-camera models

The strongest arm of Table 2 needs one trained checkpoint per camera. Whether that is a genuine cost or an implementation detail depends on whether a foreign checkpoint actually fails, which we measured across eight cameras.

It fails, and how it fails is the finding. A foreign checkpoint scores a clean frame 2.13× higher than the correct checkpoint does (median 0.163 against 0.077), and its anomalous-minus-clean margin collapses from $+0.017 \ldots + 0.061$ to a median of $+0.0009$ — a factor of 41. The distinction matters more than any AUROC: a model that were merely *offset* could be rescued by a per-camera threshold, and this one is not offset. On an unfamiliar camera, "this view is unfamiliar" dominates the reconstruction error and drowns "this patch is anomalous". The cause is the viewpoint and not the illumination: the four `3333` cameras were filmed in the same two sessions minutes apart, under identical light, and transfer *within* that family still fails. One nuance is worth preserving: on the only column with 176 pairs, foreign checkpoints still reach AUROC 0.665–0.909, so transfer preserves a weak ordering while destroying the operating point — exactly the failure a deployed system cares about and exactly the one an AUROC-only report would hide.

We then tested the authors' own answer to this objection. Dinomaly2 [19] introduces context-aware recentering (CAR), which subtracts a global summary of the image from every reconstruction target — precisely the global, view-dependent term our measurement implicates. We ported its four elements and trained eight further checkpoints on identical data, crops, schedule and hold-out, then ablated CAR alone. Table 7 attributes the effect cleanly. Because CAR normalises the reconstruction targets, absolute score levels are not comparable across variants and only the ratios are; every entry in the first two columns is a ratio for that reason.

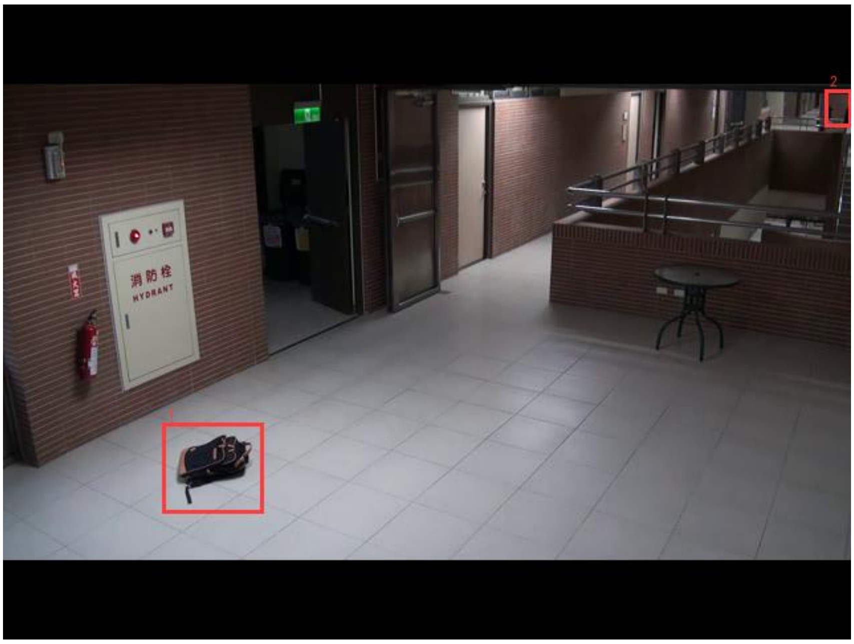

Figure 3: Proposal stage on ABODA, qualitative. The reference is the median of the opening frames and the person veto is disabled, as on CDnet. Once the people have left, the abandoned bag is the only proposal.

Without CAR the margin collapse is 43×, statistically indistinguishable from v1's 41×; with it, 10×. The loose loss, the bottleneck shape and the reversed decoder pairing do nothing at all for cross-camera behaviour — what they fix is the diagonal, where `3333-cam53` goes from 0.750 to 1.000 with CAR disabled. The mechanism proposed above is therefore confirmed by removing exactly the term it implicates and watching the effect disappear. CAR is also what costs the home camera: 0.903 without it, essentially v1's 0.920, against 0.801 with it. Recentering subtracts a global term carrying nuisance *and* signal, and on the camera where we have the most data that trade is negative.

The honest summary is a trade-off rather than an improvement: context-aware recentering buys a fourfold better cross-camera margin and pays about 0.10 image-level AUROC at home, and even the improved margin leaves transfer at chance. Per-camera training is therefore a measured requirement of this design, not a cost we chose — and to our knowledge it is not reported by the works citing either Dinomaly or AnomalyDINO, whose benchmarks contain no notion of a viewpoint that a model could fail to transfer across.

5.6 Qualitative check on abandoned objects

ABODA is the one public dataset whose anomaly class is ours, and it ships no annotations, so Figure 3 is a figure and not a number. It shows the proposal stage on an outdoor night sequence it was never tuned for: while people are present the stage proposes them, and once they leave, the bag they left behind is proposed alone. The behaviour is the one the design intends in a domain — outdoor, night, crowded — that shares nothing with a tram interior except the observation geometry.

6 Discussion

6.1 The boundary of our own claim

Object recall is pinned at 55/73 and no proposal stage moves it. Eighteen instances are localized and then rejected, four of them by the feature proposal stage alone — that is, the proposal stage found them and the judge said no. This is the exact boundary of the claim this paper makes: the proposal stage sets the ceiling, and the remaining headroom sits at the judge, where Table 4 shows prompt and model selection to be exhausted for this class of local model. The next lever is therefore adaptation — parameter-efficient fine-tuning of the judge on human-labelled crops from this deployment — which is consistent with the enabling role that Borodin et al. [12] report for PEFT in a neighbouring task. We consider that the natural continuation of this work rather than an afterthought.

6.2 Threats to validity

Dataset scale and provenance. Sixty-eight cases, 31 of them anomalous, from a single deployment, with eleven cases carrying inpainted rather than naturally occurring anomalies. Two of the four anomaly types are represented by six and seven instances respectively. Nothing in Section 5.1 should be read as a precise operating point; what it supports is a comparison *within* the table, where all four arms see exactly the same 68 cases and the same ground truth. The public-data evaluation of Section 5.4 is the mitigation, and it is a partial one: it validates the component carrying the claim, on 5 000-plus annotated frames we did not build, but it cannot validate the pipeline end to end because CDnet has no notion of a semantic verdict.

The dataset cannot be released. The footage is operational CCTV of a public-transport interior and contains identifiable passengers. This is a real limitation of the work and the reason every component-level claim in this paper is also measured on public data. It is not, however, a peculiarity of this project: the application-oriented review of in-vehicle monitoring by Caetano et al. [33] reports that work in this domain remains "fully dependent on the availability of private datasets", and identifies the extension of outdoor abandoned-object detection to vehicle interiors as an opportunity the accessible literature has overlooked. Their proposed remedy for the scarcity — synthetic augmentation — is the family our eleven inpainted cases belong to, and we regard building a releasable in-vehicle benchmark as the field-level work this paper cannot do on its own.

A single judge family carries the end-to-end numbers. Table 2 uses one judge across all four arms by construction — that is what makes it a controlled comparison — so the invariance of object recall is established for that judge. Table 4 shows that the three alternative model families reach 39–64 instances over a fixed set of boxes, which bounds but does not eliminate the concern that the invariance is model-specific.

Statistical power of the per-camera AUROCs. Four of the five columns in Table 5 and Table 7 rest on eight image pairs or fewer, where a single swapped pair moves the AUROC by 0.125. We report them for completeness and draw conclusions only from the 176-pair column and from the score-level statistics, which pool many more measurements than the AUROCs do.

6.3 What the deployment requires

One model per camera. Section 5.5 establishes this as a measured requirement rather than an implementation choice: a foreign checkpoint is blind rather than miscalibrated, so no per-camera threshold rescues it, and the authors' own remedy — context-aware recentering — halves the gap while leaving transfer at chance and costing about 0.10 AUROC on the home camera. For a fleet, the consequence is a per-camera training step of a few minutes on nominal footage, which is operationally acceptable but must be planned for. It also means that a camera whose aim is adjusted needs retraining, not merely rethresholding.

The reference must stay valid. This is the single assumption on which everything rests, and CDnet quantifies what happens when it breaks: `tramstop` at F_1 0.138 and `winterDriveway` at 0.143 are sequences whose reference goes stale under drifting outdoor light. In a tram the natural refresh point is the end-of-service inspection, when the vehicle is known to be empty. A system that cannot guarantee that refresh should expect the proposal stage to degrade toward those numbers rather than toward the 0.978 of `streetLight`.

The strongest proposal stage is strongest conditionally. The feature proposal stage wins decisively on our fleet and fails on CDnet (Section 5.4), for a reason we could isolate: it assumes a fixed camera whose reference shares its grid, and it quantises boxes to a 24 px patch. That is a good trade when the anomaly spans a substantial fraction of a patch and a bad one on low-resolution outdoor footage evaluated against pixel-accurate masks. Practitioners should read our recommendation as conditional on the deployment geometry, not as a general ranking of proposal mechanisms.

6.4 Prompting, stated carefully

Our sweep shows a prompt moving an under-firing model from 39/73 to 48/73 at unchanged specificity and moving a calibrated model not at all. We read that as calibration rather than capability, *at this scale, for these models, and without adaptation.* We deliberately do not generalise further: VERA [10] obtains real gains by treating prompts as learnable parameters, which is a different intervention from selecting among hand-written formulations, and the prompt-sensitivity literature surveyed in [25] reports the effect varying widely across tasks and model scales. What our result does support is a practical rule — measure the YES rate before attributing a recall difference to a prompt — and a direction: once a model is calibrated, further formulation work has no headroom to exploit.

6.5 What we could not do

The event-level protocol of Luna et al. [5] would let us report per-object precision, recall and F_1 on abandoned-object sequences, and is the natural complement to Section 5.4. We were unable to obtain the temporal annotation package it requires, and ABODA ships none of its own, which is why Section 5.6 is qualitative. This is a missing measurement, not a missing capability: the evaluation code exists and the sequences are public.

7 Conclusion

We presented a reference-based anomaly-detection pipeline for fixed in-vehicle CCTV in which a region-proposal stage and a local vision–language judge are independently selectable components over a shared box contract, and we used that separation to sweep each stage while the other was frozen. With the judge fixed, replacing a photometric proposal stage by a DINOv2 feature-based one raised frame-level F_1 from 0.931 to 0.984 and region precision from 0.694 to 0.896 while object recall stayed at 55/73 in every arm; with the boxes fixed, twelve model $\times$ prompt arms failed to improve on the shipped judge at usable specificity. In this pipeline the accuracy ceiling is set by the spatial proposal stage rather than by the language model, and a filter placed after the proposal stage can only ever reduce cost, never raise quality — a distinction that the current training-free vision–language anomaly pipelines cannot express, because they decide temporally over whole frames.

Two negative results are reported alongside, both measured rather than assumed: standard industrial baselines already reach 0.835–0.920 image-level AUROC on our main camera, so frame-level detection is not where a contribution can be made here; and a per-camera reconstruction model does not transfer between viewpoints, with context-aware recentering halving the gap while leaving transfer at chance and costing 0.10 AUROC at home. On CDnet 2014 the proposal stage recovers 77–100 % of annotated foreground across seven perturbation categories, with its two failure categories lying outside the in-vehicle deployment and its weakest in-domain result — illumination change — stated explicitly.

Future work follows directly from the boundary of the claim. Eighteen instances are proposed and then rejected by the judge, and prompt and model selection are exhausted at this scale, so the next lever is parameter-efficient adaptation of the judge on labelled crops from the deployment. On the evaluation side, the event-level protocol of Luna et al. [5] on public abandoned-object sequences is the missing measurement, and it needs annotations rather than method development.

Author Contributions

Conceptualization, —; methodology, —; software, —; validation, —; investigation, —; writing—original draft preparation, —; writing—review and editing, —; supervision, —. All authors have read and agreed to the published version of the manuscript.

Funding

This research was supported by VSB – Technical University of Ostrava, grant number —.

Institutional Review Board Statement

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Data Availability Statement

The tram footage analysed in this study is operational CCTV of a public-transport interior containing identifiable passengers and cannot be released. The evaluation code, the ground-truth annotation format, the per-run reports and the decision log are available in the project repository. The public datasets used are CDnet 2014 [13] and ABODA, both available from their respective authors.

Conflicts of Interest

The authors declare no conflict of interest.

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